COMPUTER NETWORKS — PYQ-STYLE PRACTICE PAPER 2

MODULE 1 + MODULE 2 | 50 MARKS | 90 MINUTES

This paper is newly generated after reviewing the uploaded 12-paper PYQ ZIP. It deliberately follows the recurring style: case studies, topology calculations, end-to-end delay, switching comparison, CRC/Hamming/checksum and scenario-based justification.

Q	Module	PYQ-style focus	Marks
1	Module 1	Case study — hierarchical topology + OSI mapping	10
2	Module 1	Topology numerical + full-mesh comparison	10
3	Module 2	End-to-end delay numerical + switching comparison	10
4	Module 2	CRC numerical + error detection	10
5	Module 2	Case study — VC/datagram + Hamming/checksum	10

QUESTION PAPER

Q1. Case Study — Hierarchical Topology and OSI Model [10 Marks]

A university is building a five-floor academic block. Each floor has 18 computers. On every floor, computers are grouped into three laboratories. " "The university wants low cost at the access level, easy troubleshooting, and a reliable backbone between floors. " "Each floor must communicate with every other floor, and the administration server is located on Floor 1.

" "**(a)** Select a suitable topology for computers inside each laboratory and justify it. [2]
" "**(b)** Select a suitable topology for connecting the floor-level networks to a central backbone. Justify your choice. [2]
" "**(c)** If the university demands maximum redundancy between the backbone nodes, which topology could be used? Give one disadvantage. [2]
" "**(d)** Explain the journey of a message from a computer on Floor 5 to the administration server on Floor 1 using the OSI model. Mention the main role of every layer. [4]

Q2. Topology Numerical and Design Decision [10 Marks]

A research centre has **10 nodes**. Compare a fully interconnected mesh with a star network. " "In the star network, each node is 20 m from the central switch. In the mesh network, the average direct cable distance between any pair of nodes is 20 m. " "Cable costs ■60/m and the central switch costs ■18,000.

" "**(a)** For full mesh, calculate the total number of links and the number of I/O ports required at each node. [4]
" "**(b)** Calculate the total cable length and installation cost of the star network. [2]
" "**(c)** Calculate the total cable length and installation cost of the full-mesh network. [2]
" "**(d)** If the research centre gives first priority to fault tolerance and second priority to cost, which topology would you recommend? Justify. [2]

Q3. End-to-End Delay Numerical [10 Marks]

A packet of **12,000 bits** travels from Host A to a server through three links. " "The propagation speed is **2.4 × 10■ m/s**. Ignore queuing and processing delays.

" "**Link 1:** distance = 5 km, rate = 20 Mbps
" "**Link 2:** distance = 800 km, rate = 100 Mbps
" "**Link 3:** distance = 10 km, rate = 50 Mbps

" "**(a)** Calculate the transmission delay on each link. [3]
" "**(b)** Calculate the propagation delay on each link. [3]
" "**(c)** Find the total end-to-end delay. [2]
" "**(d)** If the packet size is doubled while rates and distances remain unchanged, what happens to transmission delay and propagation delay? [2]

Q4. CRC Numerical [10 Marks]

A communication system uses the generator polynomial **G(x) = x■ + x + 1**. " "The data sequence is **101100101**.

" "**(a)** Write the generator in binary and determine the number of CRC check bits. [2]
" "**(b)** Perform modulo-2 division at the transmitter and find the CRC remainder. [4]
" "**(c)** Construct the transmitted codeword. [2]
" "**(d)** Explain how the receiver detects an error if one bit of the received codeword is flipped. [2]

Q5. Case Study — Packet Switching, Hamming and Checksum [10 Marks]

A disaster-response network connects mobile teams to a control centre. Sensor messages are short and bursty, while a command channel carries " "continuous two-way voice/video. Some packets must remain in order because the receiving device has no mechanism for reordering packets.

" "**(a)** For the bursty sensor traffic, choose between datagram packet switching and circuit switching. Give two reasons. [2]
" "**(b)** For the command channel where packet order must be preserved without relying on an application-level reordering mechanism, choose between datagram and virtual-circuit packet switching. Explain the setup/data/teardown phases. [3]
" "**(c)** A 4-bit dataword 1011 is encoded using even-parity Hamming code. Determine the number of parity bits required and explain where they are placed. [2]
" "**(d)** During transmission, one bit is flipped. Explain how the Hamming syndrome identifies the erroneous position and how correction is performed. [2]
" "**(e)** State the receiver's expected result when a valid Internet checksum is added to the received data using one's-complement arithmetic. [1]

DETAILED SOLUTIONS

Solution 1 — Hierarchical Topology + OSI

(a) Star is suitable inside each laboratory: every computer connects to a central switch. It is easy to troubleshoot and expand; one end-device link failure normally affects only that device.

" "**(b)** A **hierarchical/tree topology** is suitable for connecting floor-level networks to the backbone. It naturally supports access → distribution/backbone hierarchy and makes management easier.

" "**(c) Full mesh** gives maximum path redundancy because each backbone node has direct links to all others. Its disadvantage is cost/complexity: number of links grows as n(n–1)/2.

" "**(d)** Message journey: Application — user/application service; Presentation — translation, encryption/compression when required; " "Session — establishes/manages sessions; Transport — segmentation and end-to-end delivery; Network — logical addressing and routing; " "Data Link — framing, MAC addressing and hop-to-hop delivery; Physical — transmission of bits/signals. " "At the receiver the process is reversed through decapsulation.

Solution 2 — Topology Numerical

Given n = 10.

" "**(a) Full mesh:**
" "Links = n(n–1)/2 = 10×9/2 = **45 links**.
" "Each node needs n–1 = **9 I/O ports**.

" "**(b) Star:**
" "Links = n = 10.
Cable = 10×20 = **200 m**.
Cost = 200×■60 + ■18,000 = **■30,000**.

" "**(c) Full mesh:**
" "Cable = 45×20 = **900 m**.
Cost = 900×■60 = **■54,000**.

" "**(d)** If fault tolerance is the first priority, **full mesh** is the stronger choice because many alternate paths exist. " "If cost dominates, star is much cheaper. This is the key case-study trade-off.

Solution 3 — End-to-End Delay

Formulae:
Transmission delay: Tt = L/R
Propagation delay: Tp = d/v

" "Given L = 12,000 bits and v = 2.4×10■ m/s.

" "**(a) Transmission delays:**
" "Link 1: 12,000/(20×10■) = **0.6 ms**.
" "Link 2: 12,000/(100×10■) = **0.12 ms**.
" "Link 3: 12,000/(50×10■) = **0.24 ms**.
" "Total transmission = **0.96 ms**.

" "**(b) Propagation delays:**
" "5 km = 5,000 m → 5000/(2.4×10■) = **0.02083 ms**.
" "800 km = 800,000 m → **3.33333 ms**.
" "10 km = 10,000 m → **0.04167 ms**.
" "Total propagation = **3.39583 ms**.

" "**(c)** Total end-to-end delay = 0.96 + 3.39583 = **4.35583 ms**.

" "**(d)** Doubling packet size doubles every transmission delay because Tt = L/R. Propagation delay does **not** change because it depends on distance and propagation speed.

Solution 4 — CRC

(a) G(x)=x■+x+1 → binary generator = **10011**. Degree r=4, so **4 check bits** are generated.

" "**(b)** Append four zeros to the data:
1011001010000

" "Modulo-2 division by 10011 gives the remainder **1111**.

" "**(c)** Codeword = original data + remainder = **1011001011111**.

" "**(d)** At the receiver, divide the received codeword by 10011 using modulo-2 arithmetic. " "If the remainder is **0000**, no error is detected. If the remainder is non-zero, an error is detected. " "Flipping one bit changes the polynomial so the valid codeword is no longer divisible by the generator.

Solution 5 — Switching + Hamming + Checksum

(a) For short, bursty sensor messages from many teams, choose **datagram packet switching**. " "Network capacity is shared dynamically and no dedicated circuit is reserved during idle periods.

" "**(b)** Choose **virtual-circuit packet switching** when packet order must be preserved by the network path. " "**Setup:** a logical path is established and switching-table/VC entries are created. " "**Data transfer:** packets follow the established logical path, normally preserving order. " "**Teardown:** the VC is released and table entries are removed.

" "**(c)** For m=4 data bits, choose r such that 2^r ≥ m+r+1. " "r=3 gives 8 ≥ 4+3+1, so **3 parity bits** are required. Total codeword length = 7. " "Parity bits are placed at positions **1, 2 and 4**; data bits occupy 3,5,6,7.

" "**(d)** The receiver calculates parity checks and forms a binary syndrome. The syndrome value gives the **position of the erroneous bit**. " "For example, syndrome 101■ = 5 means bit position 5 is wrong; flip that bit to correct the codeword. " "A valid codeword gives syndrome 000.

" "**(e)** With one's-complement Internet checksum, the receiver expects the sum of all data words plus checksum to be **FFFF** " "(equivalently, its one's complement is 0000).

LAST-MINUTE PYQ FORMULA + PATTERN SHEET

Topic	Formula / exam trigger
Full mesh links	$n(n-1)/2$
Full mesh ports per node	$n-1$
Star links	n
Ring links	n
Transmission delay	$T_t = L/R$
Propagation delay	$T_p = d/v$
Total end-to-end delay	Sum of transmission + propagation (+ processing/queuing if given)
Circuit switching	Setup → data transfer → teardown
Datagram	No setup; packets can take different routes; may arrive out of order
Virtual circuit	Setup → data transfer on logical path → teardown
CRC	Append r zeros; modulo-2 divide; remainder = CRC
CRC error check	Remainder 0 → no error detected; non-zero → error detected
Hamming parity bits	$2^r \geq m+r+1$
Hamming syndrome	Syndrome gives erroneous bit position
Internet checksum	1's-complement sum + end-around carry; checksum = complement
Checksum receiver	Valid result = FFFF (or complement = 0000)
Case clue: every pair connected	Full mesh
Case clue: easy management + low cost	Star
Case clue: bursty many users	Packet switching
Case clue: dedicated path	Circuit switching
Case clue: packets must stay in order	Virtual circuit

Exam strategy: In case-study questions, write the selected concept first and then connect it directly to the requirement. In numericals, always write the formula, convert km→m and Mbps/Kbps→bps, substitute, and write the unit in the final answer. The uploaded PYQs repeatedly test exactly this style of reasoning.